

# Lab 4 Report - Istio Getting Started

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|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Student                | 姜丞骏                                                                                                                                                                                       |
| Experiment             | Lab 4: Istio install, Gateway API, Bookinfo, and dashboard                                                                                                                                |
| Execution              | GitHub Actions Ubuntu runner with Docker, minikube, kubectl and istioctl                                                                                                                  |
| Evidence repository    | <a href="https://github.com/whatokk/cloud-labs-03-04-kubernetes-istio">https://github.com/whatokk/cloud-labs-03-04-kubernetes-istio</a>                                                   |
| Successful Actions run | <a href="https://github.com/whatokk/cloud-labs-03-04-kubernetes-istio/actions/runs/27429016540">https://github.com/whatokk/cloud-labs-03-04-kubernetes-istio/actions/runs/27429016540</a> |
| Evidence log           | D:\论文\期末作业\计算机\Lab3_Lab4_Kubernetes_Istio\evidence\lab3-lab4-evidence\lab3-lab4-run.log                                                                                                   |

## Objective

Follow the Istio Getting Started guide. The experiment installs Istio, installs Gateway API CRDs, deploys the Bookinfo sample, configures Gateway API routing, accesses the Product Page, and installs dashboard add-ons including Kiali, Prometheus, Grafana, and tracing services.

## Major Steps and Results

1. Downloaded Istio 1.27.1 and installed the demo profile. 2. Verified istiod, ingress gateway, and egress gateway pods in istio-system. 3. Installed Gateway API CRDs and applied the Bookinfo Gateway/HTTPRoute. 4. Deployed Bookinfo services and confirmed all productpage, details, ratings, and reviews pods were running with Istio sidecars. 5. Accessed productpage through port-forward and verified the HTML title Simple Bookstore App. 6. Installed dashboard add-ons and verified Kiali, Prometheus, Grafana, tracing, and jaeger-collector services.

Screenshot 1. Istio download, istioctl version, demo profile installation, and namespace injection label.

[illegible]

Screenshot 2. Bookinfo deployment, Gateway/HTTPRoute creation, and product page verification.

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Lab 4 - Bookinfo
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## kubectl apply -f istio-1.27.1/samples/bookinfo/platform/kube/bookinfo.yaml
service/details created
serviceaccount/bookinfo-details created
deployment.apps/details-v1 created
service/ratings created
serviceaccount/bookinfo-ratings created
deployment.apps/ratings-v1 created
service/reviews created
serviceaccount/bookinfo-reviews created
deployment.apps/reviews-v1 created
deployment.apps/reviews-v2 created
deployment.apps/reviews-v3 created
service/productpage created
serviceaccount/bookinfo-productpage created
deployment.apps/productpage-v1 created

## kubectl wait --for=condition=available --timeout=300s deployment/productpage-v1
deployment.apps/productpage-v1 condition met

## kubectl wait --for=condition=available --timeout=300s deployment/details-v1
deployment.apps/details-v1 condition met

## kubectl wait --for=condition=available --timeout=300s deployment/ratings-v1
deployment.apps/ratings-v1 condition met

## kubectl wait --for=condition=available --timeout=300s deployment/reviews-v1
deployment.apps/reviews-v1 condition met

## kubectl wait --for=condition=available --timeout=300s deployment/reviews-v2
deployment.apps/reviews-v2 condition met

## kubectl wait --for=condition=available --timeout=300s deployment/reviews-v3
deployment.apps/reviews-v3 condition met

## kubectl get pods,services -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP              NODE
pod/details-v1-6cc9f5cc44-wxzdq    2/2     Running   0           24s   10.244.0.15     minikube
pod/hello-minikube-6fcb694474-xrcxt 1/1     Running   0           64s   10.244.0.3      minikube
pod/kubernetes-bootcamp-67fbd6b79-l586m 1/1     Running   0           44s   10.244.0.11     minikube
pod/kubernetes-bootcamp-67fbd6b79-pjx5d 1/1     Running   0           45s   10.244.0.10     minikube
pod/productpage-v1-7f885b46fc-6lvng   2/2     Running   0           24s   10.244.0.20     minikube
pod/ratings-v1-77b8b6df5b-fr9xd      2/2     Running   0           24s   10.244.0.17     minikube
pod/reviews-v1-fdbf79cd8-b7fvq       2/2     Running   0           24s   10.244.0.16     minikube
pod/reviews-v2-674c6d8b4-4hrng       2/2     Running   0           24s   10.244.0.18     minikube
pod/reviews-v3-7b775c7568-x5wq5     2/2     Running   0           24s   10.244.0.19     minikube

NAME                                TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE   SELECTOR
service/details                     ClusterIP      10.111.10.90    <none>            9080/TCP          24s   app=details
service/hello-minikube-minikube    NodePort       10.107.197.131 <none>            8080:32338/TCP    69s   app=hello-
service/kubernetes                  ClusterIP      10.96.0.1       <none>            443/TCP           71s   <none>
service/kubernetes-bootcamp         NodePort       10.98.194.50    <none>            8080:30191/TCP    54s   <none>
service/productpage                  ClusterIP      10.109.222.249 <none>            9080/TCP           24s   <none>
service/ratings                      ClusterIP      10.106.151.41   <none>            9080/TCP           24s   app=ratings
service/reviews                     ClusterIP      10.105.132.66   <none>            9080/TCP           24s   app=reviews

## kubectl apply -f istio-1.27.1/samples/bookinfo/gateway-api/bookinfo-gateway.yaml
gateway.gateway.networking.k8s.io/bookinfo-gateway created
httproute.gateway.networking.k8s.io/bookinfo created
error: timed out waiting for the condition on gateways/bookinfo-gateway

## kubectl get gateway,httproute -o wide
NAME                                CLASS      ADDRESS          PROGRAMMED   AGE
gateway.gateway.networking.k8s.io/bookinfo-gateway  istio      <none>           False        60s

NAME                                HOSTNAMES    AGE
httproute.gateway.networking.k8s.io/bookinfo         <none>       60s

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Lab 4 - Access Bookinfo
=====
Forwarding from 0.0.0.0:9080 -> 9080
Handling connection for 9080
Bookinfo productpage title:
<title>Simple Bookstore App</title>
=====

```

### Screenshot 3. Dashboard add-ons and service verification.

#### Lab 4 - Dashboard Add-ons

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## kubectl apply -f istio-1.27.1/samples/addons
serviceaccount/grafana created
configmap/grafana created
service/grafana created
deployment.apps/grafana created
configmap/istio-grafana-dashboards created
configmap/istio-services-grafana-dashboards created
deployment.apps/jaeger created
service/tracing created
service/zipkin created
service/jaeger-collector created
serviceaccount/kiali created
configmap/kiali created
clusterrole.rbac.authorization.k8s.io/kiali created
clusterrolebinding.rbac.authorization.k8s.io/kiali created
service/kiali created
deployment.apps/kiali created
serviceaccount/loki created
configmap/loki created
configmap/loki-runtime created
clusterrole.rbac.authorization.k8s.io/loki-clusterrole created
clusterrolebinding.rbac.authorization.k8s.io/loki-clusterrolebinding created
service/loki-memberlist created
service/loki-headless created
service/loki created
statefulset.apps/loki created
serviceaccount/prometheus created
configmap/prometheus created
clusterrole.rbac.authorization.k8s.io/prometheus created
clusterrolebinding.rbac.authorization.k8s.io/prometheus created
service/prometheus created
deployment.apps/prometheus created

## kubectl get pods -n istio-system -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP             NODE
NOMINATED NODE   READINESS GATES
grafana-665dcbcb9b-4vmgk            0/1     Running   0           20s   10.244.0.22    minikube
istio-egressgateway-5fbf54b4bd-p82n9 1/1     Running   0           2m4s   10.244.0.14    minikube
istio-ingressgateway-77c8b58566-v282r 1/1     Running   0           2m4s   10.244.0.13    minikube
istiod-5dc9d7b78-dqts5              1/1     Running   0           2m10s  10.244.0.12    minikube
jaeger-75846c549c-c55tj              1/1     Running   0           20s    10.244.0.23    minikube
kiali-7d8fcc45dd-6zw5x               0/1     Running   0           20s    10.244.0.24    minikube
loki-0                               0/2     ContainerCreating 0           20s    <none>          minikube
prometheus-759d7bcd86-f9xpp           0/2     ContainerCreating 0           20s    <none>          minikube

## kubectl rollout status deployment/kiali -n istio-system --timeout=240s
Waiting for deployment "kiali" rollout to finish: 0 of 1 updated replicas are available...
deployment "kiali" successfully rolled out

## kubectl rollout status deployment/prometheus -n istio-system --timeout=240s
deployment "prometheus" successfully rolled out

## kubectl get svc -n istio-system kiali prometheus grafana tracing jaeger-collector -o wide
NAME                                TYPE        CLUSTER-IP      EXTERNAL-IP      PORT(S)
AGE   SELECTOR
kiali                                ClusterIP    10.96.217.242    <none>            20001/TCP,9090/TCP
51s   app.kubernetes.io/instance=kiali,app.kubernetes.io/name=kiali
prometheus                           ClusterIP    10.103.67.108    <none>            9090/TCP
51s   app.kubernetes.io/component=server,app.kubernetes.io/instance=prometheus,app.kubernetes.io/name=prometheus
grafana                              ClusterIP    10.102.216.210    <none>            3000/TCP
51s   app.kubernetes.io/instance=grafana,app.kubernetes.io/name=grafana
tracing                              ClusterIP    10.105.151.84     <none>            80/TCP,16685/TCP
51s   app=jaeger
jaeger-collector                     ClusterIP    10.105.60.249     <none>            14268/TCP,14250/TCP,9411/TCP,4317/TCP,4318/TCP
51s   app=jaeger
```

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Screenshot 4. Final cluster state and Istio proxy status.

| Final State                                                                |                                            |              |                |             |                |           |               |
|----------------------------------------------------------------------------|--------------------------------------------|--------------|----------------|-------------|----------------|-----------|---------------|
| =====                                                                      |                                            |              |                |             |                |           |               |
| ## kubectl get all -A                                                      |                                            |              |                |             |                |           |               |
| NAMESPACE                                                                  | NAME                                       | READY        | STATUS         | RESTARTS    | AGE            |           |               |
| default                                                                    | pod/bookinfo-gateway-istio-b6f96bf84-f9jzk | 1/1          | Running        | 0           | 2m             |           |               |
| default                                                                    | pod/details-v1-6cc9f5cc44-wxzd             | 2/2          | Running        | 0           | 2m25s          |           |               |
| default                                                                    | pod/hello-minikube-6fcb694474-xrcxt        | 1/1          | Running        | 0           | 3m5s           |           |               |
| default                                                                    | pod/kubernetes-bootcamp-67fbd6b79-l586m    | 1/1          | Running        | 0           | 2m45s          |           |               |
| default                                                                    | pod/kubernetes-bootcamp-67fbd6b79-pjx5d    | 1/1          | Running        | 0           | 2m46s          |           |               |
| default                                                                    | pod/productpage-v1-7f885b46fc-6lvng        | 2/2          | Running        | 0           | 2m25s          |           |               |
| default                                                                    | pod/ratings-v1-77b8b6df5b-fr9xd            | 2/2          | Running        | 0           | 2m25s          |           |               |
| default                                                                    | pod/reviews-v1-fdbf79cd8-b7fv              | 2/2          | Running        | 0           | 2m25s          |           |               |
| default                                                                    | pod/reviews-v2-674c6d8b4-4hrng             | 2/2          | Running        | 0           | 2m25s          |           |               |
| default                                                                    | pod/reviews-v3-7b775c7568-x5wq5            | 2/2          | Running        | 0           | 2m25s          |           |               |
| istio-system                                                               | pod/grafana-665dc9cb9b-4vmgk               | 1/1          | Running        | 0           | 51s            |           |               |
| istio-system                                                               | pod/istio-egressgateway-5fbf54b4bd-p82n9   | 1/1          | Running        | 0           | 2m35s          |           |               |
| istio-system                                                               | pod/istio-ingressgateway-77c8b58566-v282r  | 1/1          | Running        | 0           | 2m35s          |           |               |
| istio-system                                                               | pod/istiod-5dc9d7b78-dqts5                 | 1/1          | Running        | 0           | 2m41s          |           |               |
| istio-system                                                               | pod/jaeger-75846c549c-c55tj                | 1/1          | Running        | 0           | 51s            |           |               |
| istio-system                                                               | pod/kiali-7d8fcc45dd-6zw5x                 | 1/1          | Running        | 0           | 51s            |           |               |
| istio-system                                                               | pod/loki-0                                 | 1/2          | Running        | 0           | 51s            |           |               |
| istio-system                                                               | pod/prometheus-759d7bcd86-f9xpp            | 2/2          | Running        | 0           | 51s            |           |               |
| kube-system                                                                | pod/coredns-7d764666f9-b9ttq               | 1/1          | Running        | 0           | 3m5s           |           |               |
| kube-system                                                                | pod/etcdd-minikube                         | 1/1          | Running        | 0           | 3m11s          |           |               |
| kube-system                                                                | pod/kube-apiserver-minikube                | 1/1          | Running        | 0           | 3m11s          |           |               |
| kube-system                                                                | pod/kube-controller-manager-minikube       | 1/1          | Running        | 0           | 3m11s          |           |               |
| kube-system                                                                | pod/kube-proxy-z9wt6                       | 1/1          | Running        | 0           | 3m6s           |           |               |
| kube-system                                                                | pod/kube-scheduler-minikube                | 1/1          | Running        | 0           | 3m11s          |           |               |
| kube-system                                                                | pod/storage-provisioner                    | 1/1          | Running        | 0           | 3m10s          |           |               |
| NAMESPACE                                                                  | NAME                                       | TYPE         | CLUSTER-IP     | EXTERNAL-IP | PORT(S)        |           |               |
| default                                                                    | service/bookinfo-gateway-istio             | LoadBalancer | 10.103.22.194  | <pending>   |                |           |               |
| 15021:31726/TCP,80:32135/TCP                                               | service/details                            | ClusterIP    | 10.111.10.90   | <none>      | 9080/TCP       |           |               |
| 2m25s                                                                      | service/hello-minikube                     | NodePort     | 10.107.197.131 | <none>      | 8080:32338/TCP |           |               |
| 3m10s                                                                      | service/kubernetes                         | ClusterIP    | 10.96.0.1      | <none>      | 443/TCP        |           |               |
| 3m12s                                                                      | service/kubernetes-bootcamp                | NodePort     | 10.98.194.50   | <none>      | 8080:30191/TCP |           |               |
| 2m55s                                                                      | service/productpage                        | ClusterIP    | 10.109.222.249 | <none>      | 9080/TCP       |           |               |
| 2m25s                                                                      | service/ratings                            | ClusterIP    | 10.106.151.41  | <none>      | 9080/TCP       |           |               |
| 2m25s                                                                      | service/reviews                            | ClusterIP    | 10.105.132.66  | <none>      | 9080/TCP       |           |               |
| 2m25s                                                                      | service/grafana                            | ClusterIP    | 10.102.216.210 | <none>      | 3000/TCP       |           |               |
| 51s                                                                        | service/istio-egressgateway                | ClusterIP    | 10.110.12.8    | <none>      | 80/TCP,443/TCP |           |               |
| 2m35s                                                                      | service/istio-ingressgateway               | LoadBalancer | 10.100.32.26   | <pending>   |                |           |               |
| 15021:32667/TCP,80:31641/TCP,443:32665/TCP,31400:30435/TCP,15443:31213/TCP | service/istiod                             | ClusterIP    | 10.110.66.182  | <none>      |                |           |               |
| 2m35s                                                                      | service/istio-egressgateway                | ClusterIP    | 10.110.12.8    | <none>      | 80/TCP,443/TCP |           |               |
| 2m41s                                                                      | service/jaeger-collector                   | ClusterIP    | 10.105.60.249  | <none>      |                |           |               |
| 14268/TCP,14250/TCP,9411/TCP,4317/TCP,4318/TCP                             | service/kiali                              | ClusterIP    | 10.96.217.242  | <none>      |                |           |               |
| 51s                                                                        | service/loki                               | ClusterIP    | 10.107.148.161 | <none>      |                |           |               |
| 20001/TCP,9090/TCP                                                         | service/loki-headless                      | ClusterIP    | None           | <none>      | 3100/TCP       |           |               |
| 51s                                                                        | service/loki-memberlist                    | ClusterIP    | None           | <none>      | 7946/TCP       |           |               |
| 51s                                                                        | service/prometheus                         | ClusterIP    | 10.103.67.108  | <none>      | 9090/TCP       |           |               |
| 51s                                                                        | service/tracing                            | ClusterIP    | 10.105.151.84  | <none>      |                |           |               |
| 80/TCP,16685/TCP                                                           | service/zipkin                             | ClusterIP    | 10.98.157.122  | <none>      | 9411/TCP       |           |               |
| 51s                                                                        | service/kube-dns                           | ClusterIP    | 10.96.0.10     | <none>      |                |           |               |
| 53/UDP,53/TCP,9153/TCP                                                     |                                            |              |                |             |                |           |               |
| NAMESPACE                                                                  | NAME                                       | DESIRED      | CURRENT        | READY       | UP-TO-DATE     | AVAILABLE | NODE SELECTOR |
| AGE                                                                        |                                            |              |                |             |                |           |               |

## Notes

Gateway API resources were created successfully. In the GitHub Actions minikube environment the gateway Programmed condition stayed False because no external load balancer address was assigned. The application access test therefore used `kubect` port-forward to the `productpage` service, which returned the expected Bookinfo HTML page.

## References

Istio official Getting Started guide: [install](#), [Gateway API](#), [Bookinfo](#), [IP/access](#), and [dashboard](#) sections.